

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 33****Topics covered :**

1. Laplace Transforms and Differential Equations
2. Stability

- The Laplace transform is a powerful tool for solving differential equations, particularly linear ordinary differential equations (ODEs) with constant coefficients. It converts differential equations into algebraic equations in the Laplace domain, which are easier to solve. The inverse Laplace transform then allows us to return to the time domain.
- Some of the common signals and their Laplace transform is listed in Table 1. If these signals correspond to the

Table 1: Laplace transform of some of the common signals

Signal $x(t)$	Laplace Transform $X(s)$	Region of Convergence (ROC)
$\delta(t)$ (Dirac delta)	1	Entire s -plane
$u(t)$	$\frac{1}{s}$	$\Re(s) > 0$
$e^{at}u(t)$	$\frac{1}{s-a}$	$\Re(s) > a$
$tu(t)$	$\frac{1}{s^2}$	$\Re(s) > 0$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\Re(s) > 0$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$	$\Re(s) > 0$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$	$\Re(s) > 0$

impulse response of a system, it can be classified as

- **Stable:** If the impulse response $h(t)$ satisfies $\lim_{t \rightarrow \infty} h(t) = 0$, then the system is stable.
- **Unstable:** If $h(t)$ grows unbounded as $t \rightarrow \infty$, the system is unstable.
- **Marginally Stable:** If $h(t)$ neither decays to zero nor grows unbounded but remains oscillatory, the system is marginally stable.

In the s -domain, the stability of a system can be assessed based on the location of the poles of $H(s)$. Some of the common impulse responses and the location of the associated poles are outlined in Table 2. An observation into Table 2, leads us to a useful way of assessing stability based on the poles of $H(s)$. A system is:

- **Stable** if all poles are in the left half of the complex s -plane ($\Re(s) < 0$).
- **Marginally stable** if poles are on the imaginary axis but not repeated.
- **Unstable** if any pole has a positive real part ($\Re(s) > 0$) or if repeated poles exist on the imaginary axis.

Table 2: Impulse response and poles

Signal $h(t)$	Poles	Zeros	Stability
$u(t)$	$s = 0$	None	Marginally stable
$e^{at}u(t)$	$s = a$	None	Stable if $a < 0$, Unstable if $a > 0$
$tu(t)$	$s = 0$ (double pole)	None	Unstable
$t^n u(t)$	$s = 0$ (multiple pole of order $n + 1$)	None	Unstable
$\delta(t)$	None	None	Always stable
$\sin(\omega t)u(t)$	$s = \pm j\omega$	None	Marginally stable
$\cos(\omega t)u(t)$	$s = \pm j\omega$	None	Marginally stable

- **Solving ODEs Using Laplace Transform:** The process of solving ordinary differential equations (ODEs) using the Laplace transform involves three main steps:

1. **Apply the Laplace Transform:** Take the Laplace transform of both sides of the equation, ensuring that initial conditions are incorporated correctly.
2. **Solve for the Output in the s -Domain:** Rearrange the equation to solve for $Y(s)$.
3. **Use the Inverse Laplace Transform:** Convert $Y(s)$ back to the time domain using inverse Laplace transform.

For example, consider the second-order differential equation:

$$y'' + ay' + by = x(t). \quad (1)$$

Taking the Laplace transform of both sides:

$$(s^2 Y(s) - sy(0) - y'(0)) + a(sY(s) - y(0)) + bY(s) = X(s). \quad (2)$$

Rearranging for $Y(s)$:

$$Y(s) = \frac{X(s) + sy(0) + y'(0) + ay(0)}{s^2 + as + b}. \quad (3)$$

We then use the inverse Laplace transform to obtain $y(t)$.

- The transfer function of a system is defined as the Laplace transform of the output to the input when initial conditions are zero.

$$H(s) = \frac{Y(s)}{X(s)}. \quad (4)$$

This function provides valuable insights into the system's behavior in the frequency domain.